MERA: Matrix-Entropy Actor-Critic for Sequential Reranking

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Abstract

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1 Introduction

In modern recommender systems, re-ranking is the final decision layer that determines how a small displayed slate is selected and arranged from a substantially larger upstream candidate pool. Unlike upstream retrieval or coarse ranking stages, re-ranking is not merely about assigning independent relevance scores to individual items. Instead, it is a constrained list-level combinatorial decision problem: the system must decide not only *which* items from the candidate pool should enter the final slate, but also *in what order* and *at which positions* they should be exposed. This distinction is especially critical in item recommendation scenarios, where position bias, cross-item interference, platform revenue, and user experience are all jointly mediated by the final page-level arrangement. Prior studies on list-wise recommendation have similarly emphasized that the key difficulty lies in modeling correlations within the exposed list rather than predicting each item independently. In particular, KuaiSim explicitly identifies request-level recommendation as a list-wise problem, where each action corresponds to an ordered list presented to the user, and conventional point-wise evaluation is insufficient for capturing cross-item patterns and list-level effects [6].

However, in real-world systems, item re-ranking is not a one-shot static ranking problem. The way exposure is allocated at the current step influences not only immediate responses such as clicks, likes, or conversions, but also the user's future interest state, session depth, return probability, and long-term engagement. Therefore, item re-ranking is more accurately viewed as a sequential decision-making problem rather than a single-step list optimization problem. This perspective is consistent with recent research on long-term recommendation. KuaiSim distinguishes three levels of recommendation tasks: request-level list-wise recommendation, whole-session sequential recommendation, and cross-session retention optimization, thereby emphasizing that each recommendation reshapes later requests and later user feedback [6]. Likewise, DT4IER argues that optimizing only immediate feedback such as CTR is often insufficient, since such short-term signals may fail to reflect long-term user satisfaction and retention, and practical systems must jointly consider immediate engagement and long-term return behavior [3]. AURO further shows that user retention optimization naturally leads to a dynamic recommendation environment in which user behavior patterns evolve over time, making long-term recommendation a non-stationary sequential control problem [5].

Although reinforcement learning has been widely adopted for long-term optimization in recommender systems, most existing methods still do not explicitly rewrite the geometry of the ranking action itself. A common design is to let the policy output a continuous control vector or latent action, and then rely on a pretrained scoring model or a predefined ranking function to produce the final recommendation

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list. For example, AURO models the recommendation environment as a hidden-parameter MDP and uses a continuous action space together with a scoring model and ranking function to optimize user retention under non-stationary dynamics [5]. These approaches are effective for long-term reward optimization, yet the ranking decision remains an indirect consequence of a downstream ranking module, rather than a native structured action. As a result, the item–position assignment that fundamentally defines re-ranking is not treated as the primary decision object.

On the other hand, research on structured action learning has shown that permutation matrices provide a natural representation for combinatorial decisions involving matching, ranking, and ordering. In particular, Sinkhorn Policy Gradient (SPG) demonstrates that actor–critic learning can be extended to permutation-valued actions by introducing a Sinkhorn layer that outputs a continuous relaxation of permutation matrices [1]. Relatedly, Gumbel-Sinkhorn networks show that a score matrix can be perturbed by Gumbel noise, relaxed by Sinkhorn scaling, and rounded to a discrete matching with the Hungarian algorithm, yielding a reparameterized sampling view for latent permutations [4]. Importantly, however, SPG is formulated in an average-reward contextual bandit setting, and the paper explicitly notes that its derivation can be viewed as a one-step reinforcement learning formulation obtained by setting the episode horizon to one; extending the method to episodic RL is left as future work [1]. In other words, existing structured-action methods clarify how permutation-valued decisions can be relaxed and sampled, but they do not address the defining challenge of sequential re-ranking: the ranking action is not merely a permutation output for a static instance, but a control variable that shapes future user states through exposure.

This leaves a clear gap in the literature. Existing long-term recommendation methods model sequential optimization but usually do not represent ranking actions in an assignment-native way. Existing permutation-based actor–critic methods show that structured combinatorial actions are learnable, but remain in contextual bandit or one-step settings rather than sequential re-ranking environments with evolving user preferences. Consequently, an important question remains unresolved: *when a ranking action should itself be treated as a structured display-or-discard decision, and when that decision continuously affects the future user-interest distribution, how should we optimize it under a unified reinforcement learning framework?*

In this work, we argue that the action in item re-ranking should not be viewed as a set of independent item scores, nor merely as an auxiliary input to a downstream ranking function. Instead, it should be modeled directly as an item–position allocation plan on the final reranking shortlist. Because the shortlist size K is larger than the number of exposed positions N, the action must simultaneously decide which N items are displayed and which $K - N$ items are discarded. Following the transport-plan notation common in differentiable top-k formulations, we represent each re-ranking action as a rectangular partial-assignment matrix whose rows index shortlisted items and whose columns index the N display positions plus one dump slot. The policy therefore outputs a score matrix whose Sinkhorn relaxation defines a soft transport plan, while the environment-facing discrete action is sampled by Gumbel perturbation followed by Sinkhorn relaxation and Hungarian rounding. This representation preserves the combinatorial structure of reranking while bringing display competition, position effects, and discard decisions into the action space itself rather than leaving them to post-processing.

Building on this action formulation, we further cast sequential re-ranking as an actor–critic problem over rectangular transport-style ranking actions. Moreover, motivated by the geometric structure induced by Sinkhorn scaling with prescribed marginals, we introduce a maximum matrix-entropy regularization principle to stabilize exploration and optimization in this structured action space. The resulting framework unifies three ingredients that have previously remained largely separate: assignment-native action modeling for ranking, long-term user optimization with evolving request histories, and entropy-regularized control tailored to partial-assignment matrices. Rather than merely reusing Sinkhorn within reinforcement learning, our formulation reinterprets re-ranking itself as a structured control problem on a ranking transport polytope.

Our main contributions are summarized as follows:

2 Problem Formulation

We consider session-wise item reranking as a sequential interaction between the agent and users, and emphasize the list-wise, multi-behavior, and session-level nature of this interaction, as illustrated in

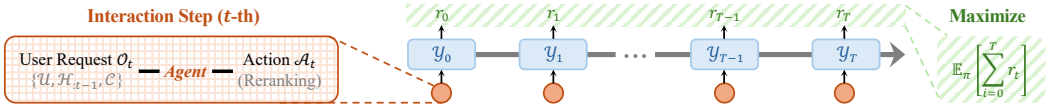

Figure 1: An overview: session-wise item reranking as a sequential interaction.

Figure 1. A user session refers to the continuous interactions from the moment the user opens the app until leaving it, and we use T to denote the terminal interaction step of that session. At the t-th interaction step in a session, the agent receives a user request $\mathcal{O}_t$ as input and outputs a structured reranking action $X_t \in \mathcal{X}_{K,N}$, which means selecting an ordered list of N items from a candidate item pool of size K. Following the standard notation in recommendation, each request $\mathcal{O}_t$ consists of static user profile features $\mathcal{U}$, the most recent interaction history $\mathcal{H}_{:t-1}$ that evolves over time, and an upstream candidate item pool $\mathcal{C}_t$. Here $K := |\mathcal{C}_t|$ is the candidate-pool size and N is the number of exposed positions in the final slate. The formal definition of $\mathcal{X}_{K,N}$ is given later in Section 3.1, Eq. (2).

After receiving X_t, the user provides a feedback signal $\mathcal{Y}_t$, which may contain multiple behavior channels and delayed outcomes, such as click, like, add-to-favorite, download, conversion, dwell, leave, or return signals. We map the multi-channel feedback to a scalar reward by $r_t = r(\mathcal{Y}_t)$ and seek a policy $\pi(\cdot|\mathcal{O}_t)$ that maximizes long-term utility:

$$\max_{\pi} \; J(\pi) := \mathbb{E}_\pi\left[\sum_{t=0}^{T} r_t\right]. \tag{1}$$

3 Matrix Entropy Reranking Actor-Critic

Our method, **M**atrix **E**ntropy **R**eranking **A**ctor-Critic **(MERA)**, is organized from structured action geometry to learning. We first represent reranking as constrained transport-style partial assignment, then introduce a compact actor–critic interface, and finally derive a practical hard-action training procedure.

3.1 Structured Action Modeling

MERA represents the final-stage reranking action as an item-position allocation on a shortlist of K items. Each item is assigned either to one of the N exposed positions or to a dump slot. Let $\mu := \mathbf{1}_K \in \mathbb{R}^K$ and $\nu := [\mathbf{1}_N^\top,\ K-N]^\top \in \mathbb{R}^{N+1}$. The corresponding hard partial-assignment set and its transport relaxation are

$$\begin{aligned} \mathcal{X}_{K,N} &:= \left\{X \in \{0,1\}^{K\times(N+1)} : X\mathbf{1}_{N+1} = \mu,\ X^\top \mathbf{1}_K = \nu\right\}, \\ \mathcal{T}_{K,N} &:= \left\{P \in \mathbb{R}_+^{K\times(N+1)} : P\mathbf{1}_{N+1} = \mu,\ P^\top \mathbf{1}_K = \nu\right\}. \end{aligned} \tag{2}$$

Any $X \in \mathcal{X}_{K,N}$ is a discrete reranking action, any $P \in \mathcal{T}_{K,N}$ is a soft exposure-allocation plan. The displayed list is read from the first N columns of the resulting assignment.

3.2 Actor and Critic Interfaces

The actor defines a parameterized policy $\pi_\phi(\cdot \mid \mathcal{O}_t)$: given request $\mathcal{O}_t$, it outputs a soft matrix $P_\phi(\mathcal{O}_t) \in \mathcal{T}_{K,N}$ as an action-distribution representation, and a sampler $\mathcal{S}$ draws a hard action by $X_t = \mathcal{S}(P_\phi(\mathcal{O}_t)) \in \mathcal{X}_{K,N}$.

The critic evaluates how good a realized action is for the current request. It takes $(\mathcal{O}_t, X_t)$ as input and outputs $Q_\theta(\mathcal{O}_t, X_t)$, which is trained to match the regularized expected return. Detailed network parameterization is deferred to Appendix A.1.

3.3 Maximum Matrix-Entropy Reinforcement Learning

Standard reinforcement learning maximizes expected discounted reward. SAC tells the same story, but adds a preference for policies with larger causal entropy $\mathcal{H}(\pi(\cdot \mid \mathcal{O}_t))$ [2]. In reranking, however,

the hard action space is the ordered top-N space $\mathcal{X}_{K,N}$ with size $|\mathcal{X}_{K,N}| = \frac{K!}{(K-N)!}$, so computing policy entropy exactly already costs $O\Big(\frac{K!}{(K-N)!}\Big)$.

MERA keeps the maximum-entropy viewpoint but replaces policy entropy with a tractable upper bound. For a hard-action policy π, let $P_\pi(\mathcal{O}_t) = \mathbb{E}[X_t \mid \mathcal{O}_t] \in \mathcal{T}_{K,N}$ be the mean transport plan and let $H_{\mathrm{mat}}(P) := -\sum_{i=1}^{K}\sum_{j=1}^{N+1} P_{ij}\log P_{ij}$. Since a ranking action can be viewed as the joint assignment of items to slots, Shannon entropy subadditivity gives $\mathcal{H}(\pi(\cdot \mid \mathcal{O}_t)) \le H_{\mathrm{mat}}(P_\pi(\mathcal{O}_t))$. We therefore optimize

$$J_\tau(\phi) = \mathbb{E}_{\pi_\phi}\left[\sum_{t=0}^{T}\gamma^t\Big(r(\mathcal{Y}_t) + \tau H_{\mathrm{mat}}(P_t)\Big)\right], \qquad P_t = P_\phi(\mathcal{O}_t). \tag{3}$$

This yields a matrix-entropy policy-evaluation operator $\mathcal{T}_\tau^\phi$ and the improvement operator

$$(\mathcal{T}_\tau Q)(\mathcal{O}_t, P_t) = \mathbb{E}_{(\mathcal{O}_{t+1},\mathcal{Y}_t)\sim p(\cdot|\mathcal{O}_t,P_t)}\left[r(\mathcal{Y}_t) + \gamma \sup_{P'\in\mathcal{T}_{K,N}}\Big(Q(\mathcal{O}_{t+1},P') + \tau H_{\mathrm{mat}}(P')\Big)\right]. \tag{4}$$

Proposition 1 (Policy evaluation contraction)**.** *For any fixed actor plan P_ϕ, the operator $\mathcal{T}_\tau^\phi$ is a γ-contraction under the sup norm and therefore admits a unique fixed point Q_τ^ϕ.*

Proposition 2 (Soft optimality contraction)**.** *The operator $\mathcal{T}_\tau$ is also a γ-contraction under the sup norm and therefore admits a unique fixed point $Q_\tau^\star$.*

These operators are the matrix-entropy analogue of SAC soft policy iteration; unlike SAC, the improvement step is a constrained optimization over transport plans rather than a KL projection onto a policy density. The omitted derivations are deferred to Appendix A.

3.4 Decoupled Optimization

In practice, the inner maximization in Eq. (4) is still too expensive to solve at every update, so we approximate the next-step regularized value with the target actor. Let $\bar{P}_{t+1} := P_{\bar{\phi}}(\mathcal{O}_{t+1})$ and $\bar{X}_{t+1} \sim \pi_{\bar{\phi}}(\cdot \mid \mathcal{O}_{t+1})$. We then use the target

$$\hat{q}_t = r(\mathcal{Y}_t) + \gamma\Big(Q_{\bar{\theta}}(\mathcal{O}_{t+1}, \bar{X}_{t+1}) + \tau H_{\mathrm{mat}}(\bar{P}_{t+1})\Big),$$

which keeps the same reward-plus-next-soft-value structure as Eq. (4). The first term evaluates the next hard action under the target critic, while the second retains the matrix-entropy bonus of the next feasible plan; target networks $(\bar{\theta}, \bar{\phi})$ stabilize this bootstrap and avoid solving the inner maximization over $\mathcal{T}_{K,N}$ explicitly. The critic therefore minimizes

$$\mathcal{L}_Q(\theta) = \mathbb{E}\Big[\big(Q_\theta(\mathcal{O}_t, X_t) - \hat{q}_t\big)^2\Big]. \tag{5}$$

Once Q_θ approximates this regularized return, the actor is updated by maximizing hard-action value under the feasible marginal-plan constraint,

$$\max_\phi\ \mathbb{E}_{\mathcal{O}\sim\mathcal{D},\ X\sim\pi_\phi(\cdot|\mathcal{O})}\Big[Q_\theta(\mathcal{O}, X) + \tau H_{\mathrm{mat}}(P_\phi(\mathcal{O}))\Big] \quad \text{s.t.} \quad P_\phi(\mathcal{O}) \in \mathcal{T}_{K,N}, \tag{6}$$

where $P_\phi(\mathcal{O})\mathbf{1}_{N+1} = \mu$ and $P_\phi(\mathcal{O})^\top\mathbf{1}_K = \nu$ impose the partial doubly-stochastic marginal constraints. To optimize Eq. (6) rigorously, we decouple its differentiable surrogate into two linked problems: an inner entropic transport layer over feasible plans, and an outer alignment problem between the actor logits and the critic-induced gradient field.

Theorem 1 (KKT-Equivalent Decoupling)**.** *Let $G_\theta(\mathcal{O}, \varepsilon)$ denote the critic gradient obtained through the straight-through sampler, and let Π remove row and column shift directions of the logits. Then solving the differentiable surrogate of Eq.* (6) *is first-order equivalent to solving the inner problem*

$$P_\phi^\star(\mathcal{O}, \varepsilon) = \arg\max_{P\in\mathcal{T}_{K,N}}\Big\{\langle Z_\phi(\mathcal{O}) + \varepsilon, P\rangle + \tau H_{\mathrm{mat}}(P)\Big\}, \tag{7}$$

and the outer alignment condition

$$\Pi\Big(Z_\phi(\mathcal{O}) + \varepsilon - G_\theta(\mathcal{O}, \varepsilon)\Big) = 0. \tag{8}$$

The inner problem in Eq. (7) is solved exactly by generalized Sinkhorn scaling.

Proposition 3 (Sinkhorn Layer)**.** *The unique solution of Eq.* (7) *is*

$$P_\phi^\star(\mathcal{O}, \varepsilon) = \mathrm{Sinkhorn}_{\mu,\nu}((Z_\phi(\mathcal{O}) + \varepsilon)/\tau)\,. \tag{9}$$

Here $\mathrm{Sinkhorn}_{\mu,\nu}$ *denotes the unique matrix in* $\mathcal{T}_{K,N}$ *obtained by scaling* $\exp((Z_\phi(\mathcal{O}) + \varepsilon)/\tau)$ *to row marginals* μ *and column marginals* ν*.*

The outer problem then becomes a logit-alignment objective, which we optimize through

$$\mathcal{L}_Z(\phi) = \frac{1}{2}\,\mathbb{E}_{\mathcal{O}\sim\mathcal{D},\ \varepsilon\sim\mathrm{Gumbel}}\left[\left\|\Pi\Big(Z_\phi(\mathcal{O}) + \varepsilon - \mathrm{sg}[G_\theta(\mathcal{O}, \varepsilon)]\Big)\right\|_F^2\right], \tag{10}$$

where $\mathcal{D}$ denotes the empirical request marginal of the replay buffer and $\mathrm{sg}[\cdot]$ denotes stop-gradient. The proofs of the KKT-equivalent decoupling, the Sinkhorn layer, and the final actor loss are deferred to Appendix A.5, Appendix A.4, and Appendix A.6, respectively.

References

[1] Patrick Emami and Sanjay Ranka. Learning permutations with sinkhorn policy gradient. *arXiv preprint arXiv:1805.07010*, 2018.

[2] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. *Proceedings of the 35th International Conference on Machine Learning*, 80:1861–1870, 2018.

[3] Ziru Liu, Shuchang Liu, Zijian Zhang, Qingpeng Cai, Xiangyu Zhao, Kesen Zhao, Lantao Hu, Peng Jiang, and Kun Gai. Sequential recommendation for optimizing both immediate feedback and long-term retention. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024.

[4] Gonzalo Mena, David Belanger, Scott Linderman, and Jasper Snoek. Learning latent permutations with gumbel-sinkhorn networks. In *International Conference on Learning Representations*, 2018.

[5] Zhenghai Xue, Qingpeng Cai, Bin Yang, Lantao Hu, Peng Jiang, Kun Gai, and Bo An. Auro: Reinforcement learning for adaptive user retention optimization in recommender systems. In *Proceedings of the ACM Web Conference 2025*, 2025.

[6] Kesen Zhao, Shuchang Liu, Qingpeng Cai, Xiangyu Zhao, Ziru Liu, Dong Zheng, Peng Jiang, and Kun Gai. Kuaisim: A comprehensive simulator for recommender systems. In *Advances in Neural Information Processing Systems*, 2023.

A Appendix: Derivations for MERA

This appendix provides the derivations omitted from the main text. We keep the main body focused on definitions and final statements, while placing the intermediate steps and proofs here.

A.1 Detailed Network Parameterization

This subsection gives the explicit matrix-form parameterization used in Section 3.2. At step t, the request encoder is

$$\begin{aligned} s_t &= f_s(\mathcal{O}_t) \in \mathbb{R}^{d_s}, \\ C_t &= f_c(\mathcal{C}_t) = \left[c_{t,1}^\top, \ldots, c_{t,K}^\top\right]^\top \in \mathbb{R}^{K\times d_e}, \\ E &= \left[e_1^\top, \ldots, e_{N+1}^\top\right]^\top \in \mathbb{R}^{(N+1)\times d_e}. \end{aligned} \tag{11}$$

Here slots $1, \ldots, N$ are exposed positions and slot $N+1$ is the dump slot.

$$\begin{aligned} g_t^{(s)} &= \sigma(W_s s_t + b_s) \in \mathbb{R}^{d_e}, \qquad g_t^{(e)} = \sigma(W_e s_t + b_e) \in \mathbb{R}^{d_e}, \\ \hat{h}_{t,i} &= c_{t,i} \odot g_t^{(s)},\ i \in [K], \qquad \hat{e}_{t,j} = e_j \odot g_t^{(e)},\ j \in [N+1], \\ \hat{H}_t &= \left[\hat{h}_{t,1}^\top, \ldots, \hat{h}_{t,K}^\top\right]^\top, \qquad \widetilde{H}_t = \mathrm{Transformer}_\phi^{\mathrm{set}}(\hat{H}_t), \\ [Z_\phi(s_t)]_{ij} &= \langle \widetilde{h}_{t,i}, \hat{e}_{t,j}\rangle, \qquad Z_\phi(s_t) \in \mathbb{R}^{K\times(N+1)}. \end{aligned} \tag{12}$$

$$
\begin{aligned}
R_t &= (X_t^\top C_t)_{1:N,:}, \\
Q_\theta(s_t, X_t) &= \mathrm{MLP}_\theta\Big(\big[\, s_t;\ \mathrm{MeanPool}\big(\,\mathrm{Transformer}_\theta^{\mathrm{ord}}(R_t)\big)\,\big]\Big).
\end{aligned}
\tag{13}
$$

A.2 Proof of Proposition 1

For a fixed actor plan P_ϕ, the regularized evaluation operator is

$$
(\mathcal{T}_\tau^\phi Q)(\mathcal{O}, P) = \mathbb{E}_{(\mathcal{O}',\mathcal{Y})\sim p(\cdot|\mathcal{O},P)}\Big[r(\mathcal{Y}) + \gamma\big(Q\big(\mathcal{O}', P_\phi(\mathcal{O}')\big) + \tau H_{\mathrm{mat}}(P_\phi(\mathcal{O}'))\big)\Big]. \tag{14}
$$

For any $(\mathcal{O}, P)$, let $\Delta(\mathcal{O}, P) := (\mathcal{T}_\tau^\phi Q_1)(\mathcal{O}, P) - (\mathcal{T}_\tau^\phi Q_2)(\mathcal{O}, P)$. Then the contraction proof can be written as one displayed derivation:

$$
\begin{aligned}
\|\mathcal{T}_\tau^\phi Q_1 - \mathcal{T}_\tau^\phi Q_2\|_\infty &= \sup_{(\mathcal{O},P)} |\Delta(\mathcal{O}, P)| \\
&= \sup_{(\mathcal{O},P)} \big|(\mathcal{T}_\tau^\phi Q_1)(\mathcal{O}, P) - (\mathcal{T}_\tau^\phi Q_2)(\mathcal{O}, P)\big| \\
&= \gamma \sup_{(\mathcal{O},P)} \Big|\mathbb{E}_{(\mathcal{O}',\mathcal{Y})}\Big[Q_1\big(\mathcal{O}', P_\phi(\mathcal{O}')\big) - Q_2\big(\mathcal{O}', P_\phi(\mathcal{O}')\big)\Big]\Big| \\
&\qquad \text{(by Eq. (14), since the reward and future entropy terms cancel)} \\
&\le \gamma \sup_{(\mathcal{O},P)} \mathbb{E}_{(\mathcal{O}',\mathcal{Y})}\Big[\big|Q_1\big(\mathcal{O}', P_\phi(\mathcal{O}')\big) - Q_2\big(\mathcal{O}', P_\phi(\mathcal{O}')\big)\big|\Big] \\
&\le \gamma \|Q_1 - Q_2\|_\infty.
\end{aligned}
\tag{15}
$$

Hence $\mathcal{T}_\tau^\phi$ is a γ-contraction and admits a unique fixed point by Banach's fixed-point theorem.

A.3 Proof of Proposition 2

The regularized optimality operator is

$$
(\mathcal{T}_\tau Q)(s, P) = \mathbb{E}_{(s',\mathcal{Y})\sim p(\cdot|s,P)}\left[r(\mathcal{Y}) + \gamma \sup_{P'\in\mathcal{T}_{K,N}} \Big(Q(s', P') + \tau H_{\mathrm{mat}}(P')\Big)\right]. \tag{16}
$$

For any (s, P), let $\Delta^\star(s, P) := (\mathcal{T}_\tau Q_1)(s, P) - (\mathcal{T}_\tau Q_2)(s, P)$ and

$$
\Psi_i(s') := \sup_{P'\in\mathcal{T}_{K,N}} \Big(Q_i(s', P') + \tau H_{\mathrm{mat}}(P')\Big), \qquad i \in \{1, 2\}.
$$

Then the optimality contraction is obtained from the single multi-line derivation

$$
\begin{aligned}
\|\mathcal{T}_\tau Q_1 - \mathcal{T}_\tau Q_2\|_\infty &= \sup_{(s,P)} |\Delta^\star(s, P)| \\
&= \gamma \sup_{(s,P)} \big|\mathbb{E}_{(s',\mathcal{Y})}\big[\Psi_1(s') - \Psi_2(s')\big]\big| \\
&\qquad \text{(by Eq. (16))} \\
&\le \gamma \sup_{(s,P)} \mathbb{E}_{(s',\mathcal{Y})}\left[\sup_{P'\in\mathcal{T}_{K,N}} \big|Q_1(s', P') - Q_2(s', P')\big|\right] \\
&\le \gamma \|Q_1 - Q_2\|_\infty.
\end{aligned}
\tag{17}
$$

where the inequality uses the standard bound $|\sup f - \sup g| \le \sup |f - g|$ with $f(P') = Q_1(s', P') + \tau H_{\mathrm{mat}}(P')$ and $g(P') = Q_2(s', P') + \tau H_{\mathrm{mat}}(P')$. so $\mathcal{T}_\tau$ is also a γ-contraction and admits a unique fixed point.

A.4 Derivation of Proposition 3

The inner entropic transport problem used in Proposition 3 is

$$
P_\phi^\star(\mathcal{O}, \varepsilon) = \arg\max_{P\in\mathcal{T}_{K,N}} \Big\{\langle Z_\phi(\mathcal{O}) + \varepsilon, P\rangle + \tau H_{\mathrm{mat}}(P)\Big\}. \tag{18}
$$

To keep the derivation compact, write $B := Z_\phi(\mathcal{O}) + \varepsilon$. Let $\mathfrak{L}(P, \xi, \zeta)$ denote the Lagrangian of Eq. (18) with row and column multipliers ξ and ζ for the transport constraints with source marginal μ and target marginal ν. The first-order condition yields the multi-line derivation

$$\begin{aligned}
\mathbf{0} = \frac{\partial \mathfrak{L}}{\partial P_{ij}} &= B_{ij} - \tau(1 + \log P_{ij}) + \xi_i + \zeta_j \\
\iff \log P_{ij} &= \frac{B_{ij} + \xi_i + \zeta_j}{\tau} - 1 \\
\iff P_{ij} &= \exp\left(\frac{B_{ij} + \xi_i + \zeta_j}{\tau} - 1\right) = u_i \exp(B_{ij}/\tau) v_j \\
\iff P &= \mathrm{Diag}(u)\ \exp(B/\tau)\ \mathrm{Diag}(v) \\
\iff P_\phi^\star &= \mathrm{Sinkhorn}_{\mu,\nu}(B/\tau),
\end{aligned} \tag{19}$$

where the constant $\exp(-1)$ is absorbed into the row and column scalings, and the last line is exactly the definition of generalized Sinkhorn scaling over $\exp(B/\tau)$ with row marginals μ and column marginals ν.

A.5 Derivation of Theorem 1

The straight-through regularized objective used in the main text is

$$\mathcal{J}_\theta^{\mathrm{st}}(\mathcal{O}, \varepsilon, P) = Q_\theta\Big(\mathcal{O}, \mathrm{sg}\left(\mathcal{M}_{\mu,\nu}(P) - P\right) + P\Big) + \tau H_{\mathrm{mat}}(P), \qquad P \in \mathcal{T}_{K,N}. \tag{20}$$

The corresponding projector is

$$\Pi(M) = H_K M H_{N+1}, \qquad H_m := I_m - \frac{1}{m}\mathbf{1}_m \mathbf{1}_m^\top. \tag{21}$$

Equivalently,

$$\Pi(M) = M - \frac{1}{N+1} M \mathbf{1}_{N+1}\mathbf{1}_{N+1}^\top - \frac{1}{K}\mathbf{1}_K\mathbf{1}_K^\top M + \frac{1}{K(N+1)}\mathbf{1}_K\mathbf{1}_K^\top M \mathbf{1}_{N+1}\mathbf{1}_{N+1}^\top,$$

which removes all row-shift and column-shift directions.

Fix a request $\mathcal{O}$, a noise sample ε, and assume $P \in \mathrm{int}(\mathcal{T}_{K,N})$. Let

$$X^{\mathrm{st}}(P) := \mathrm{sg}\big(\mathcal{M}_{\mu,\nu}(P) - P\big) + P.$$

Because stop-gradient freezes $\mathcal{M}_{\mu,\nu}(P) - P$ in the backward pass, the derivative of the straight-through critic term in Eq. (20) with respect to P is exactly the gradient of Q_θ with respect to its matrix input, evaluated at $X^{\mathrm{st}}(P)$. Write

$$G_\theta(\mathcal{O}, \varepsilon; P) := \nabla_A Q_\theta(\mathcal{O}, A)|_{A = X^{\mathrm{st}}(P)}.$$

Then the KKT stationarity condition of Eq. (20) can be compressed into the projected form

$$\begin{aligned}
&\mathbf{0} = G_\theta(\mathcal{O}, \varepsilon; P) - \tau\big(\log P + \mathbf{1}_K \mathbf{1}_{N+1}^\top\big) + \xi \mathbf{1}_{N+1}^\top + \mathbf{1}_K \zeta^\top \qquad \text{(KKT stationarity in } P\text{-space)} \\
&\implies \Pi\Big(G_\theta(\mathcal{O}, \varepsilon; P) - \tau \log P\Big) = 0 \\
&\qquad \text{(because } \Pi(\xi \mathbf{1}_{N+1}^\top) = \Pi(\mathbf{1}_K \zeta^\top) = \Pi(\mathbf{1}_K \mathbf{1}_{N+1}^\top) = 0).
\end{aligned} \tag{22}$$

If moreover $P = \mathrm{Sinkhorn}_{\mu,\nu}((Z + \varepsilon)/\tau)$, then there exist row and column shift vectors α and β such that the generalized Sinkhorn scaling identity gives

$$\begin{aligned}
\log P &= (Z + \varepsilon)/\tau + \alpha \mathbf{1}_{N+1}^\top + \mathbf{1}_K \beta^\top \\
&\implies \Pi(\tau \log P) = \Pi(Z + \varepsilon) \\
&\qquad \text{(the row and column shifts are again removed by } \Pi).
\end{aligned} \tag{23}$$

Substituting this relation into Eq. (22) yields

$$\Pi\Big(G_\theta(\mathcal{O}, \varepsilon; P) - \tau \log P\Big) = 0 \iff \Pi\Big(Z + \varepsilon - G_\theta(\mathcal{O}, \varepsilon; P)\Big) = 0. \tag{24}$$

Finally, setting $P = \widetilde{P}_\phi(\mathcal{O}, \varepsilon) = \mathrm{Sinkhorn}_{\mu,\nu}((Z_\phi(\mathcal{O}) + \varepsilon)/\tau)$ gives the statement in Theorem 1, since $G_\theta(\mathcal{O}, \varepsilon; P)$ then reduces to the main-text quantity $G_\theta(\mathcal{O}, \varepsilon)$.

A.6 Why the MERA Actor Loss is a KKT Residual Minimization

Let

$$\begin{aligned}
\widetilde{P}_{\mathrm{ref}}(\mathcal{O},\varepsilon) &= \mathrm{sg}\Big[\,\mathrm{Sinkhorn}_{\mu,\nu}((Z_\phi(\mathcal{O})+\varepsilon)/\tau)\Big],\\
X_{\mathrm{ref}}(\mathcal{O},\varepsilon) &= \mathcal{M}_{\mu,\nu}\big(\widetilde{P}_{\mathrm{ref}}(\mathcal{O},\varepsilon)\big),\\
X^{\mathrm{st}}_{\mathrm{ref}}(\mathcal{O},\varepsilon) &= \mathrm{sg}\Big(X_{\mathrm{ref}}(\mathcal{O},\varepsilon)-\widetilde{P}_{\mathrm{ref}}(\mathcal{O},\varepsilon)\Big)\;+\;\widetilde{P}_{\mathrm{ref}}(\mathcal{O},\varepsilon),\\
G_\theta(\mathcal{O},\varepsilon) &= \mathrm{sg}\Big[\,\nabla_A Q_\theta(\mathcal{O},A)|_{A=X^{\mathrm{st}}_{\mathrm{ref}}(\mathcal{O},\varepsilon)}\Big].
\end{aligned}$$

Then the zero-residual condition of the actor loss in Eq. (10) can be written as

$$\begin{aligned}
\mathcal{L}_Z(\phi;\mathcal{O},\varepsilon)=0 \iff\;& \Pi\Big(Z_\phi(\mathcal{O})+\varepsilon-\mathrm{sg}[G_\theta(\mathcal{O},\varepsilon)]\Big)=0\\
\iff\;& \Pi\Big(Z_\phi(\mathcal{O})+\varepsilon-G_\theta(\mathcal{O},\varepsilon)\Big)=0\\
\iff\;& \widetilde{P}_\phi(\mathcal{O},\varepsilon)=\mathrm{Sinkhorn}_{\mu,\nu}((Z_\phi(\mathcal{O})+\varepsilon)/\tau)\\
&\text{and the first-order KKT condition of Eq. (20) holds.}
\end{aligned} \tag{25}$$

where the second equivalence uses the fact that stop-gradient changes only the backward graph, not the forward value. Importantly, this statement concerns stationarity of the straight-through objective, not necessarily local maximality. Local optimality further requires second-order sufficient conditions, or global concavity of $\mathcal{J}^{\mathrm{st}}_\theta(\mathcal{O},\varepsilon,P)$ over $\mathcal{T}_{K,N}$.

Gradient of the actor loss. Because the teacher branch is detached, the chain rule collapses to the logit branch only:

$$\begin{aligned}
\nabla_\phi\mathcal{L}_Z &= \left(\frac{\partial Z_\phi}{\partial\phi}\right)^\top\frac{\partial\mathcal{L}_Z}{\partial Z_\phi}\qquad(\text{because } \partial\widetilde{P}_{\mathrm{ref}}/\partial\phi=\partial X^{\mathrm{st}}_{\mathrm{ref}}/\partial\phi=\partial G_\theta/\partial\phi=0)\\
&= \mathbb{E}_{\mathcal{O}\sim\mathcal{D},\,\varepsilon\sim\mathrm{Gumbel}}\left[\left(\frac{\partial\Pi(Z_\phi(\mathcal{O})+\varepsilon)}{\partial\phi}\right)^\top\Pi\Big(Z_\phi(\mathcal{O})+\varepsilon-\mathrm{sg}[G_\theta(\mathcal{O},\varepsilon)]\Big)\right],
\end{aligned} \tag{26}$$

where the detached branches remove all gradients through Hungarian rounding, target matching, and the teacher critic path. Equation (26) therefore shows that MERA updates the actor through the straight-through estimator: the critic is evaluated on hard sampled actions, while the backward pass follows the relaxed Gumbel-Sinkhorn path.

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